Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY

BACHELOR OF TECHNOLOGY

COMPUTER NETWORK (303105256)

4TH SEMESTER

DEPARTMENT Of CSE-AI & DS

Laboratory Manual

DATA COMMUNICATION AND NETWORKING PRACTICAL BOOK

INFORMATION AND TECHNOLOGY DEPARTMENT

PREFACE

It gives us immense pleasure to present the first edition of Data Communication and Networking Practical Book for the B. Tech. 3rd year students for **PARUL UNIVERSITY.**

The Data Communication and Networking theory and laboratory course at **PARUL UNIVERSITY, WAGHODIA, and VADODARA** is designed in such a way that students develop the basic understanding of the subject in the theory classes and then try their hands on the experiments to realize the various devices and circuits learnt during the theoretical sessions. The main objective of the Data Communication and Networking: Learning what is a computer network, topologies and through Experimentations. All the experiments are designed to understand the characteristic of various electronics devices and electronics circuits and also to expose the students to various electronics simulators and their uses.

The objective of this Data Communication and Networking Practical Book is to provide a comprehensive source for all the experiments included in the Data Communication and Networking laboratory course. It explains all the aspects related to every experiment such as: understanding of computer networks, signal processing between different electronics circuits and practical designing of practical books.

We acknowledge the authors and publishers of all the books which we have consulted while developing this Practical book. Hopefully this Data Communication and Networking *Practical Book* will serve the purpose for which it has been developed.

INSTRUCTIONS TO STUDENTS

1. Be prompt in arriving to the laboratory and always come well prepared for the experiment.
2. Be careful while working on the equipments operated with high voltage power supply.
3. Work quietly and carefully. Give equal opportunity to all your fellow students to work on the instruments.
4. Every student should have his/her individual copy of the Data Communication and Networking Practical Book.
5. Every student has to prepare the notebooks specifically reserved for the Data Communication and Networking Practical work "Data Communication and Networking Practical Book".
6. Every student has to necessarily bring his/her Data Communication and Networking Practical Book, Data Communication and Networking Practical Class Notebook and Data Communication and Networking Practical Final Notebook, when he/she comes to the practical to perform the experiment.
7. Record your observations honestly. Never makeup reading or doctor them to get a better fit on the graph or to produce the correct result. Display all your observations on the graph (If applicable).
8. All the observations have to be neatly recorded in the Data Communication and Networking Practical Class Notebook (as explained in the Data Communication and Networking Practical Book) and verified by the instructor before leaving the laboratory.
9. If some of the experiments appear to be wrong then repeat the set of observations carefully.
10. Do not share your readings with your fellow student. Every student has to produce his/her own set of readings by performing the experiment separately.
11. After verification of the recorded observations, do the calculation in the Data Communication and Networking Practical Class Notebook (as explained in the Data Communication and Networking Practical Book) and produce the desired results and get them verified by the instructor.
12. Never forget to mention the units of the observed quantities in the observation table. After calculations, represent the results with appropriate units.
13. Calculate the percentage error in the results obtained by you if the standard results are available and also try to point out the sources of errors in the experiment.
14. Find the answers of all the questions mentioned under the section 'Find the Answers' at the end of each experiment in the Data Communication and Networking Practical Book.
15. Do not forget to get the information of your next allotment (the experiment which is to be performed by you in the next laboratory session) before leaving the laboratory from the Technical Assistant.
16. Find the answers of all the questions mentioned under the section 'Find the Answers' as the end of each experiment in the Data Communication and Networking Practical Book.
17. Finally record the verified observations along with the calculation and results in the Data Communication and Networking Practical Final Notebook.
18. Do not forget to get the information of your next allotment (the experiment which is to be performed by you in the next laboratory session) before leaving the laboratory from the Technical Assistant.
19. The grades for the Data Communication and Networking Practical course work will be awarded based on your performance in the laboratory, regularity, recording of experiments in the Data Communication and Networking Practical Final Notebook, lab quiz, regular viva-voce and end-term examination.

CERTIFICATE

This is to certify that

Mr./Ms.....

.....

..... with enrolment no. .....has successfully

completed his/her laboratory experiments in the Computer Network

(303105256) from the department

of

..... during

the academic year .....

Date of Submission:..... Staff In charge:.....

Head Of Department:.....

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2	**Experiments of Packet capture tool: Wireshark** To understand the features of wireshark as a packet capture tool and understand encapsulation of information. Also study effect of few network commands						
3	**To study behaviour of generic devices used for networking: (CISCO PACKET TRACER)** Design a simple network with multiple nodes and connect via generic devices available in library. Perform simulation and trace communication behaviour of specified network devices. 1: Use only HUB to design a small network having 4 to 6 hosts 2: Use only a Switch to design a small network with 4 to 6 hosts. 3: Use both the device (HUB and SWITCH) for a network and find out functioning difference between switch and hub.						
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8	**Implementation of SUBNETTING:** Design multiple subnet with suitable number of hosts. Make a plan to assign static IP addressing across all subnet to explain implementation of SUBNETTING.						
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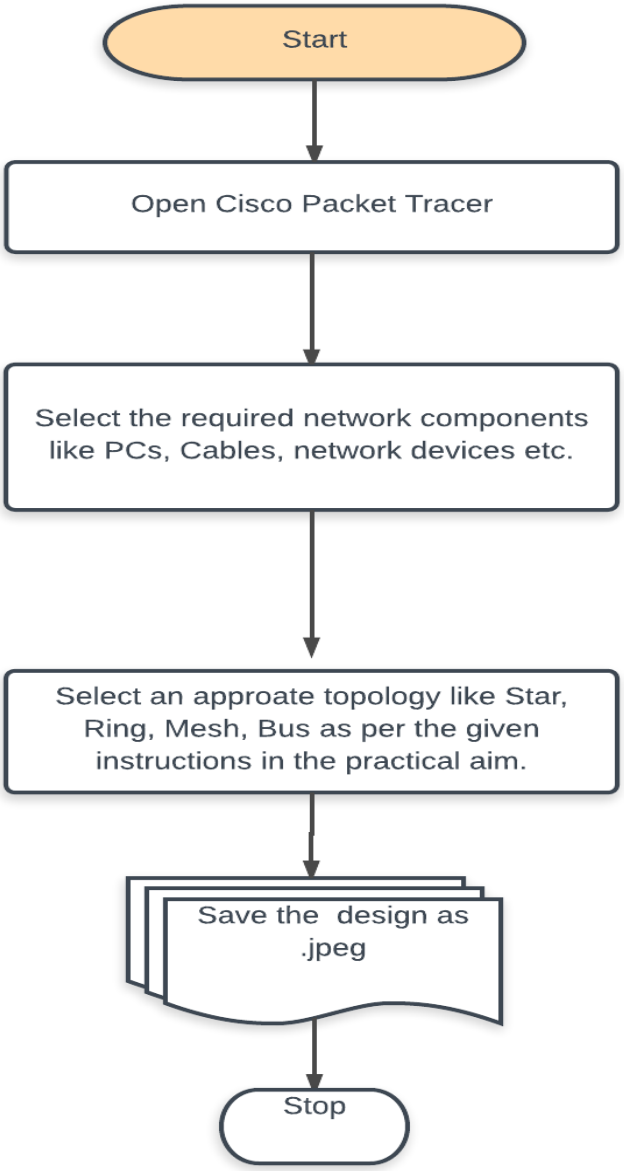
EXPERIMENT NO: 1

AIM: Experiments on Simulation Tools: (CISCO PACKET TRACER): To understand environment of CISCO PACKET TRACER to design simple network and perform experiments.

19.1 Purpose:

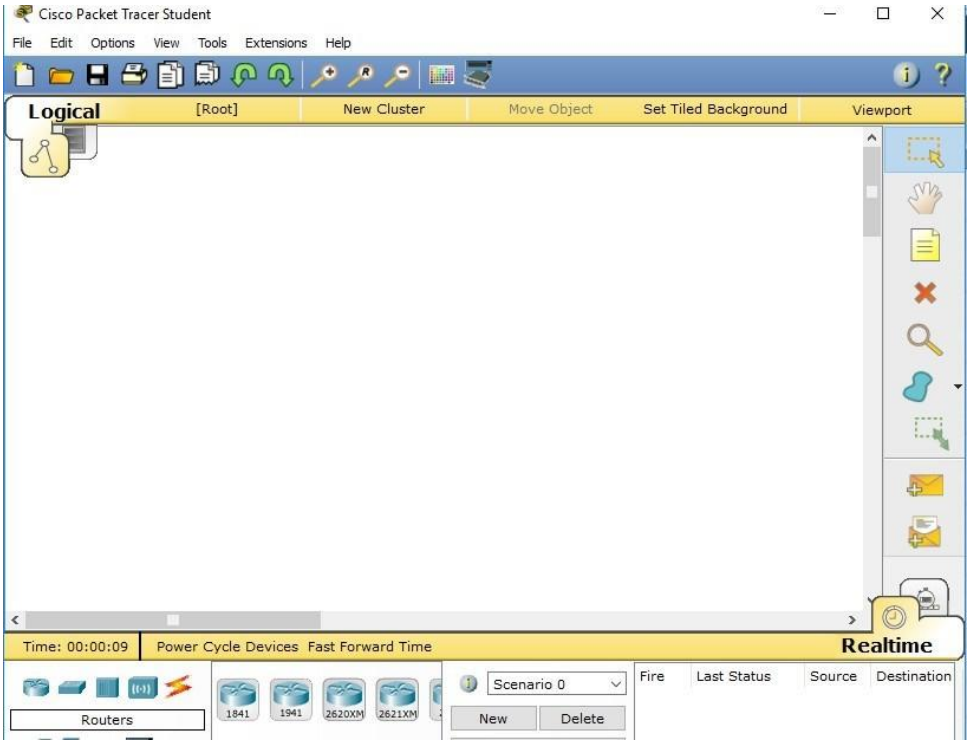
The main objective of the proposed experiment is to give exposure of numerous simulation devices to students such that student can actually design various network topologies in real-time.

19.2 Logical Flow of Practical

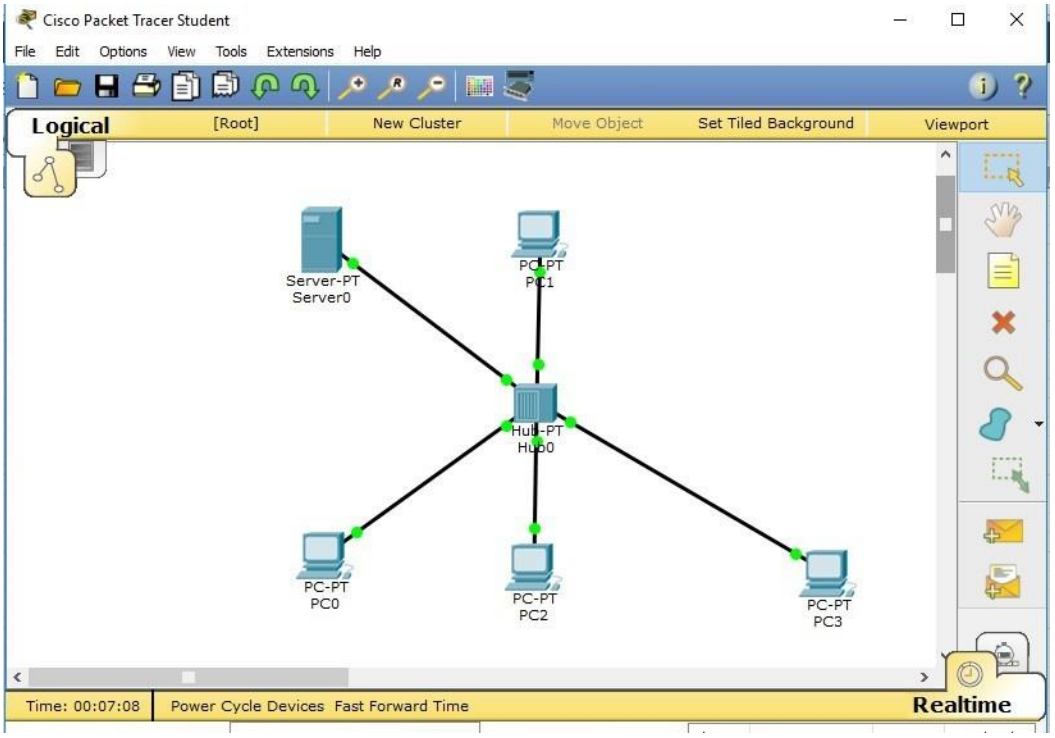


19.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



19.4 Expected Output:



19.5 Practice Excercises:

1. **Design a Bus topology network using Cisco Packet Tracer resources.**
2. **Design a Star Topology network using Cisco Packet Tracer resources.**
3. **Design a Mesh Topology network using Cisco Packet Tracer resources.**

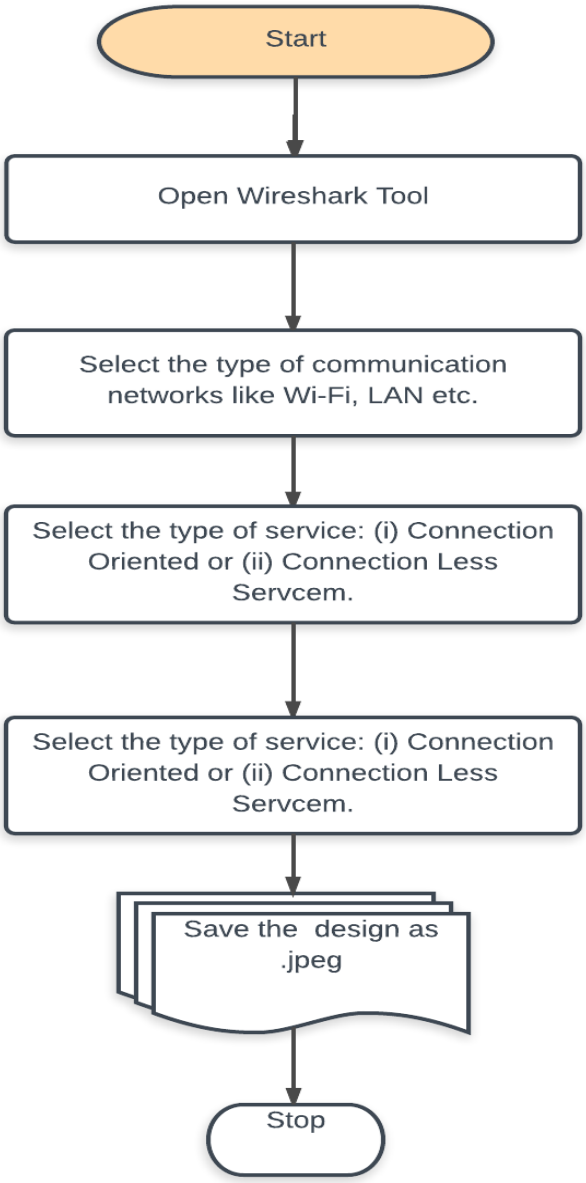
EXPERIMENT NO: 2

AIM: To understand the features of wireshark as a packet capture tool and understand encapsulation of information. Also study effect of few network commands

2.1 Purpose:

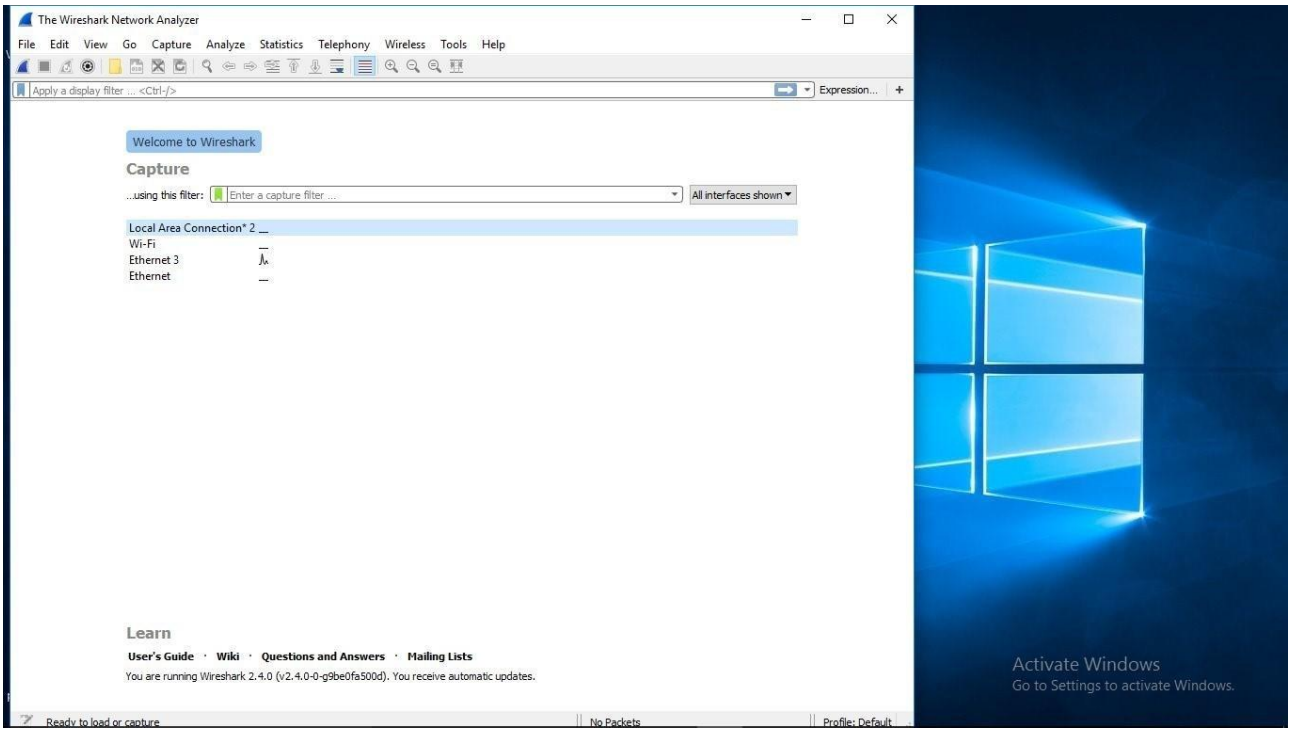
The main objective of the proposed experiment is to give exposure of wire-shark tool to students so students can learn to monitor transmission packets being sent in Wi-Fi and LAN environments.

2.2 Logical Flow of Practical



2.3 TOOLS/SOFTWARE:

Wireshark



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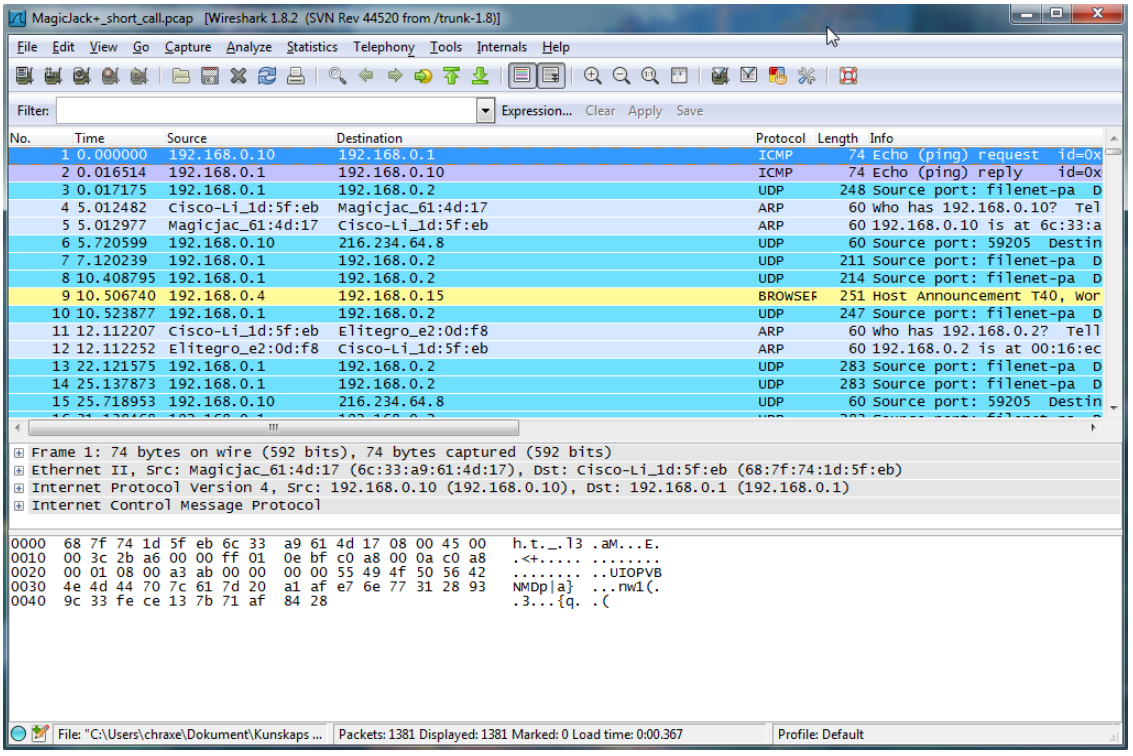


2.5 Practice Excercises:

1. Monitor TCP packets in a LAN connection and record your observations.
2. Monitor UDP packets in a LAN connection and record your observations.
3. Monitor TCP packets in a Wi-Fi connection and record your observations.
4. Monitor UDP packets in a Wi-Fi connection and record your observations.

EXPERIMENT NO: 3

AIM: To study behaviour of generic devices used for networking:

Design a simple network with multiple nodes and connect via generic devices available in library. Perform simulation and trace communication behaviour of specified network devices.

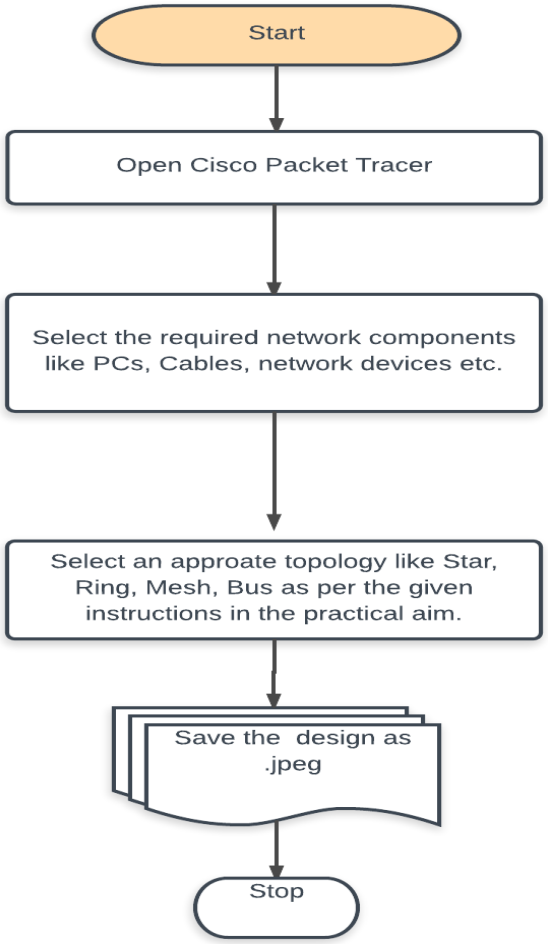
1: Use only HUB to design a small network having 4 to 6 hosts

2: Use only a Switch to design a small network with 4 to 6 hosts.

3: Use both the device (HUB and SWITCH) for a network and find out functioning difference between switch and hub.

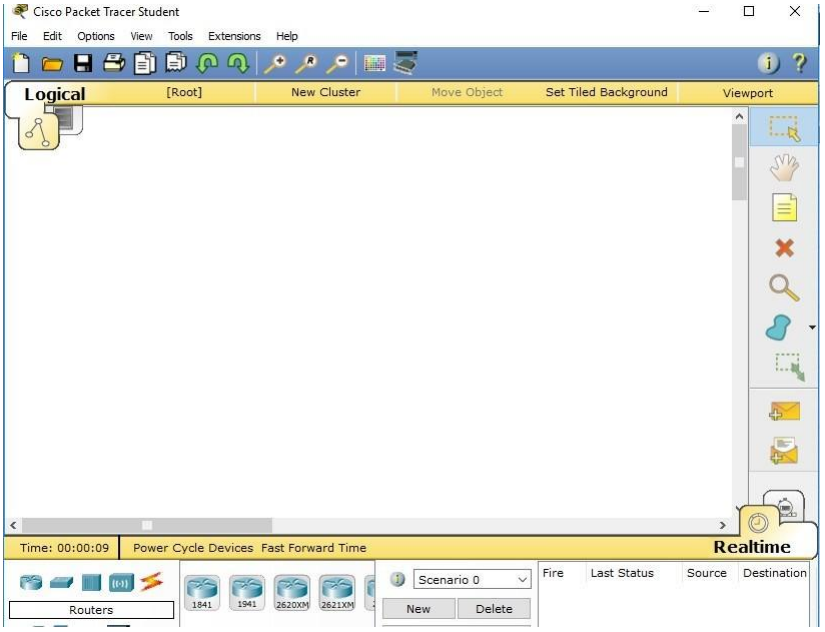
3.1 Purpose:
The main objective of the proposed experiment is to give exposure of numerous simulation devices to students such that student can actually design various network topologies in real-time using Hub and Switch network devices.

3.2 Logical Flow of Practical

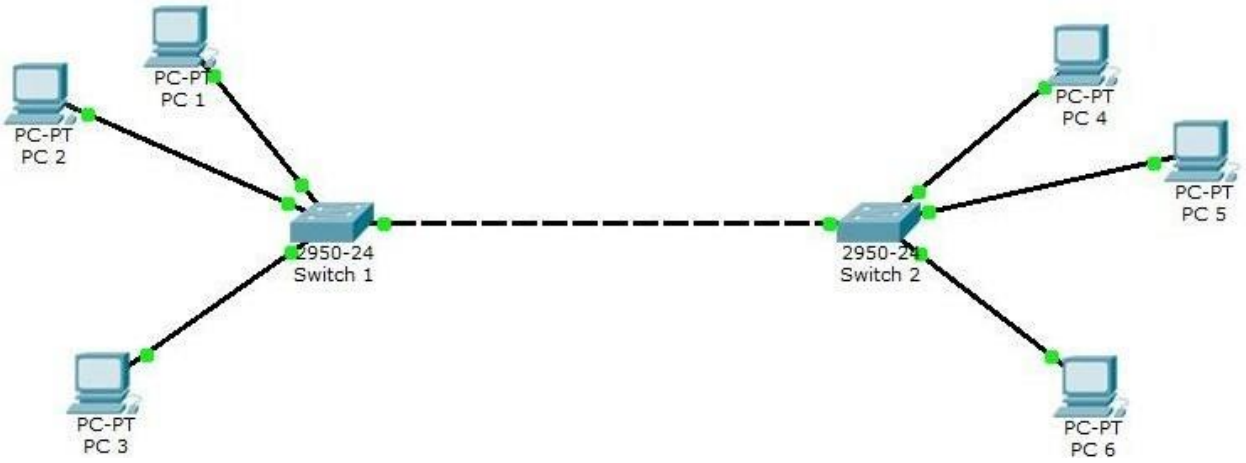


3.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



3.4 Expected Output:



3.5 Practice Excercises:

1. Design a Star and Bus topology network. Connect the designed network using multiple switches.

3. Design a Bus and Mesh Topology network using Cisco Packet Tracer resources.

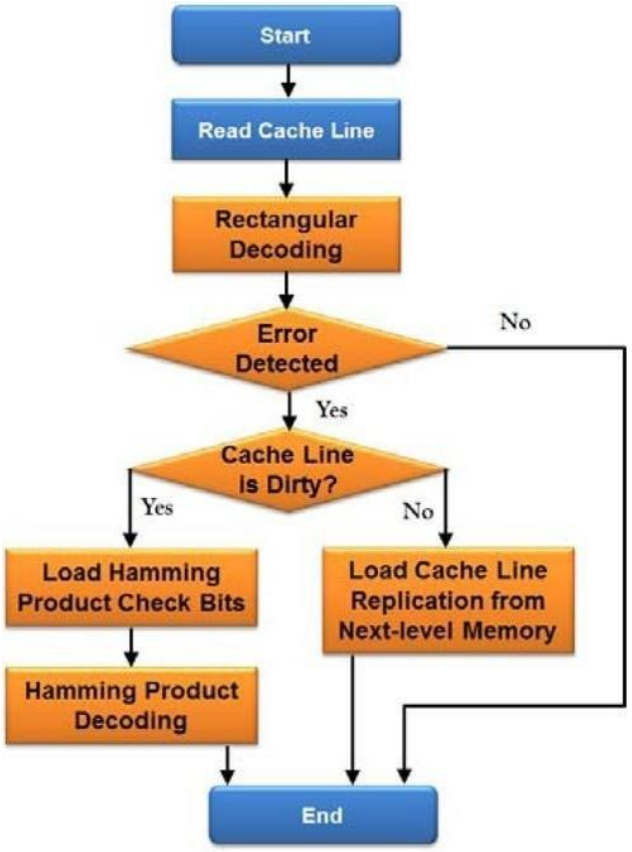
EXPERIMENT NO: 4

AIM: Data Link Layer (Error Correction): Write a program to implement error detection and correction using HAMMING code concept. Make a test rum to input data stream and verify error correction feature.

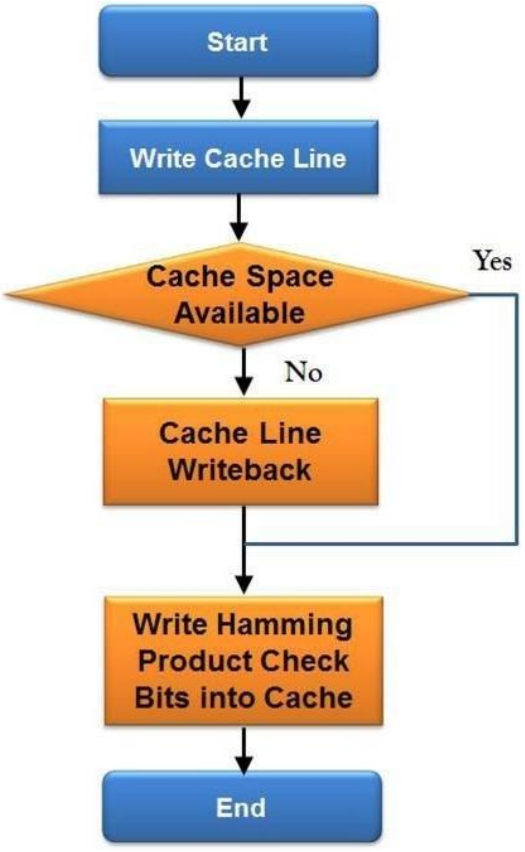
4.1 Purpose:

The main objective of the proposed experiment is to understand error correction between a sender and a receiver using a hamming code concept.

4.2 Logical Flow of Practical



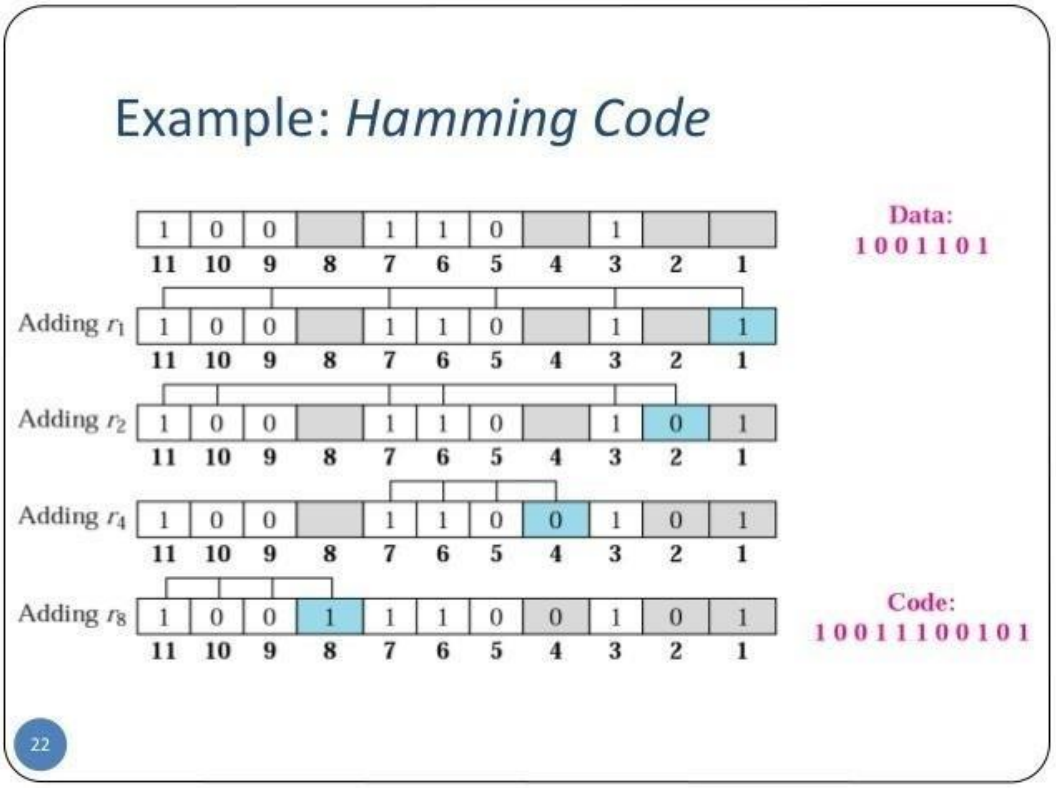
(a)



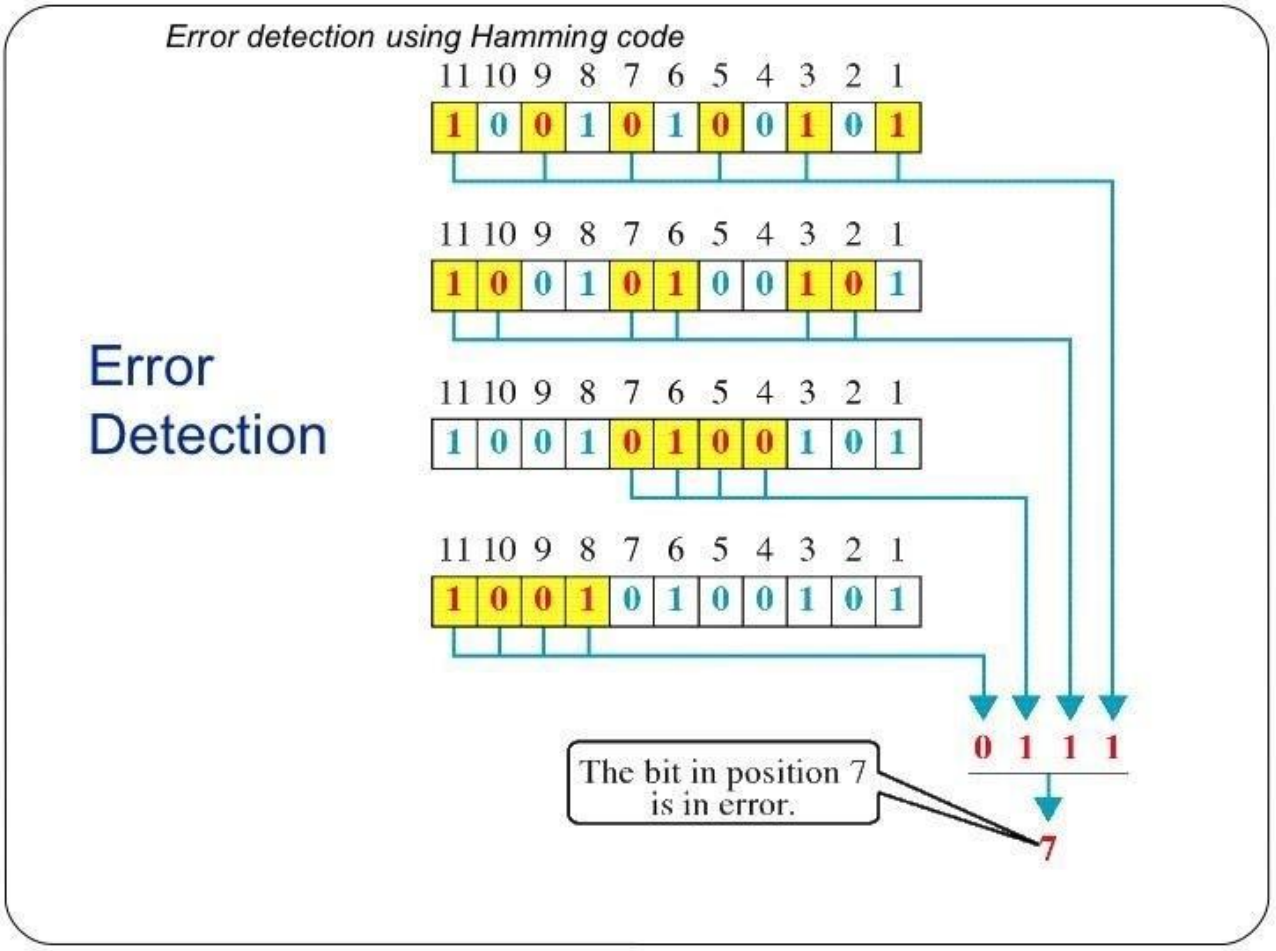
(b)

4.3 TOOLS/SOFTWARE:

Language C



4.4 Expected Output:



4.5 Practice Excercises:

1. Apply hamming code error correction technique in any packet switching network.

2. Apply hamming code error correction technique in any circuit switching network.

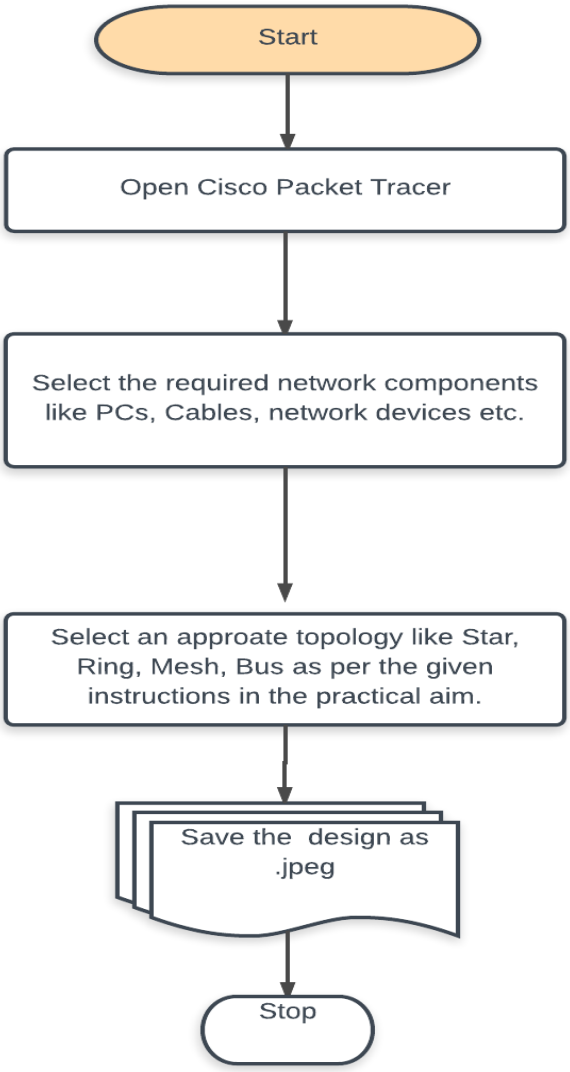
EXPERIMENT NO: 5

AIM: Virtual LAN: Simulate Virtual LAN configuration using CISCO Packet Tracer Simulation.

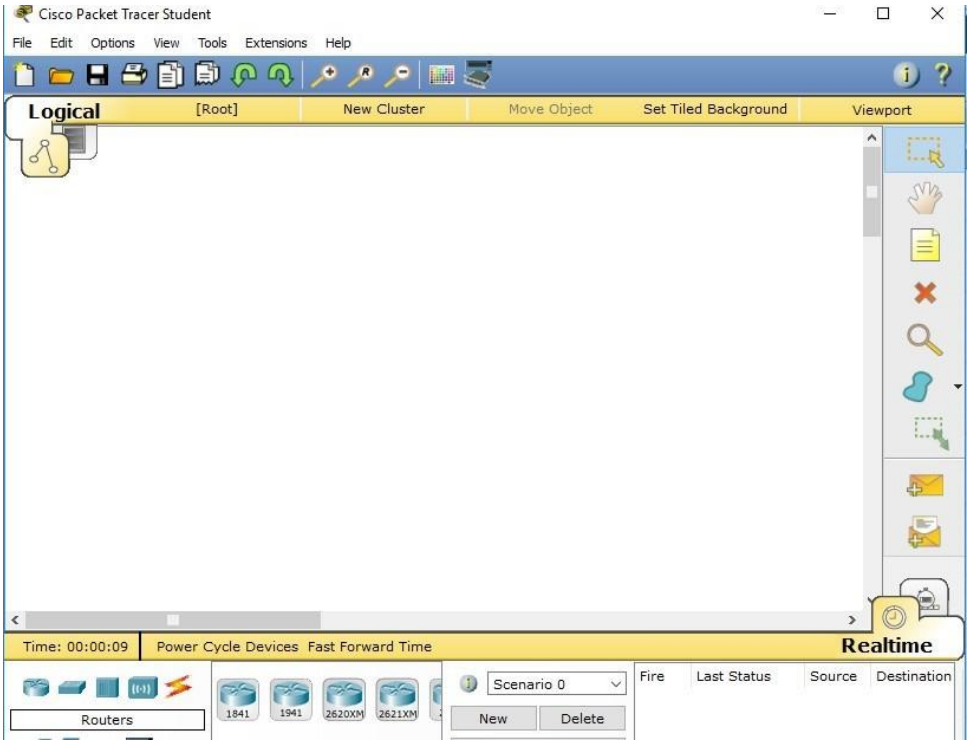
5.1 Purpose:

The main objective of the proposed experiment is to establish a virtual private network using Cisco Packet Tracer resources and learn various LAN settings. Also design various wireless network topologies in real-time using Hub and Switch network devices.

5.2 Logical Flow of Practical



5.3 TOOLS/SOFTWARE:
Cisco Packet Tracer



5.4 Expected Output:

5.5 Practice Excercises:

1. Design a Wireless Star and Bus topology network. Connect the designed network using multiple switches.

2. Design a Wireless Bus and Mesh Topology network using an access point.

EXPERIMENT NO: 6

AIM: Wireless LAN: Configuration of Wireless LAN using CISCO Packet Tracer

6.1 Purpose:

The main objective of the proposed experiment is to establish wireless LAN network using Cisco Packet Tracer resources, learn various LAN settings and simulation network using Router and Switch network devices.

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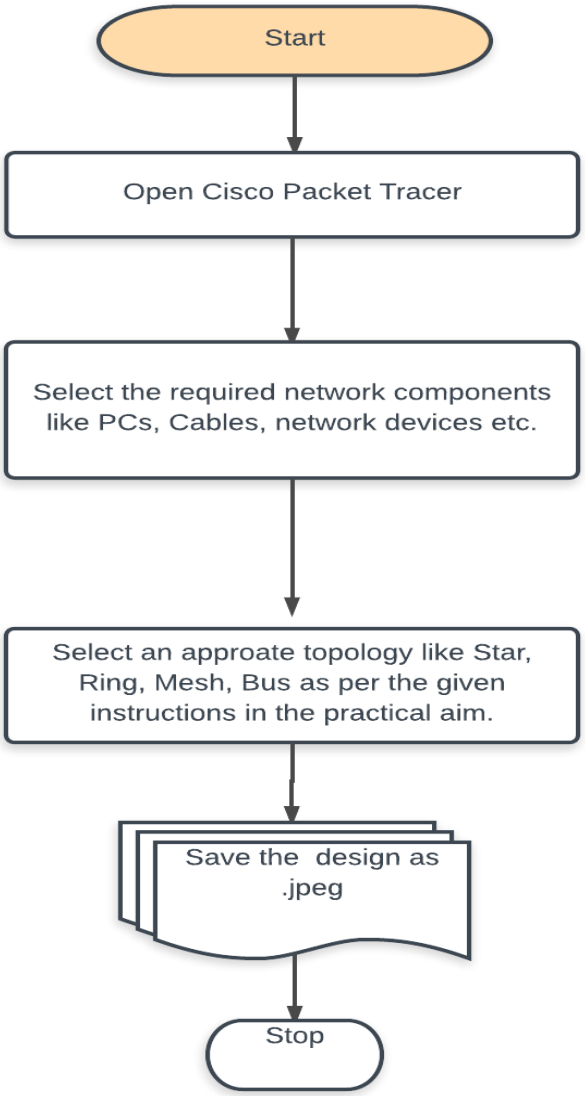
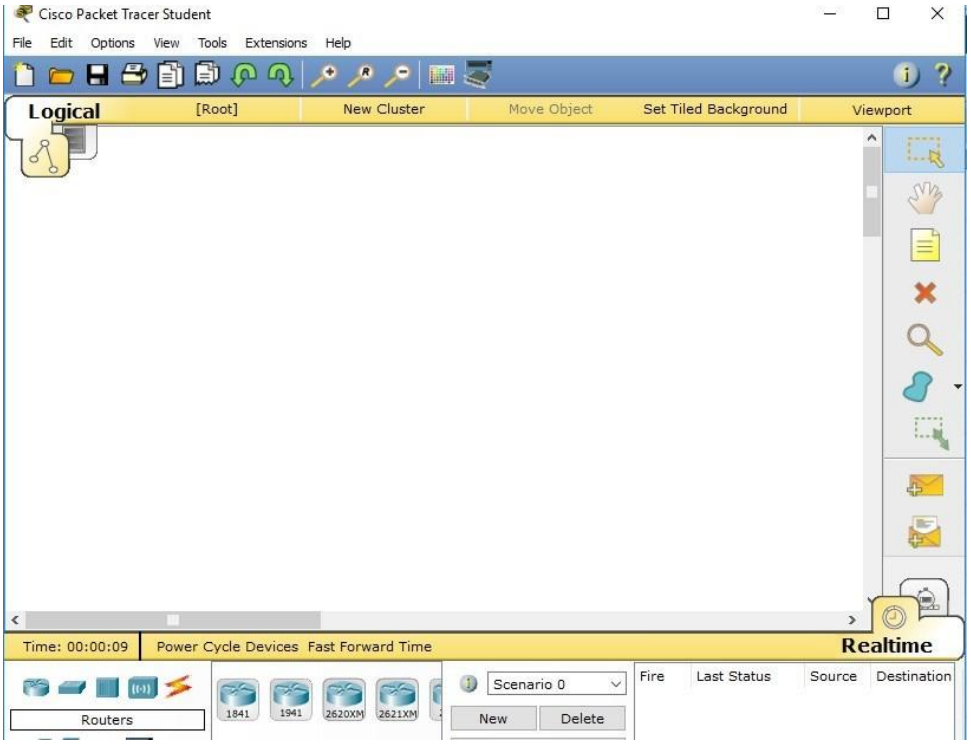
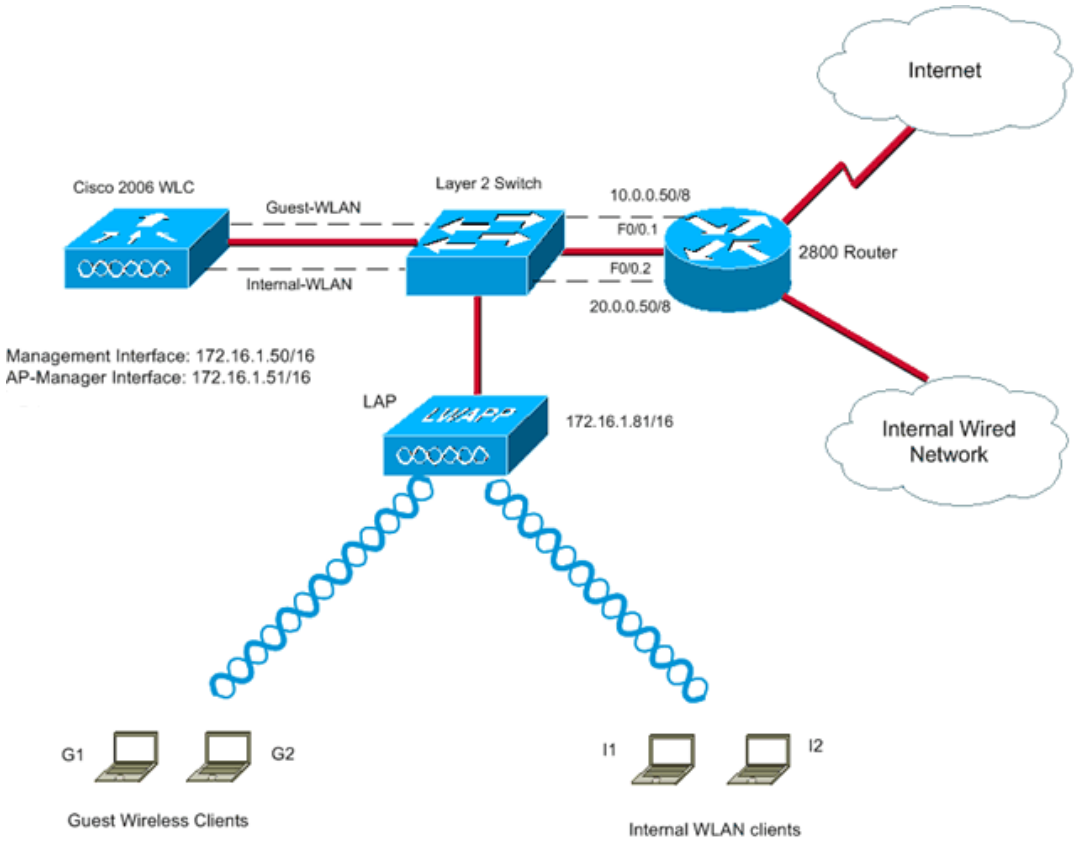
6.2 Logical Flow of Practical



6.3 TOOLS/SOFTWARE:
Cisco Packet Tracer



6.4 Expected Output:



6.5 Practice Excercises:

1. Design a Wireless LAN using a wireless router, nodes and access point.

2. Design a VPN network using an access point and a wireless router.

EXPERIMENT NO: 7

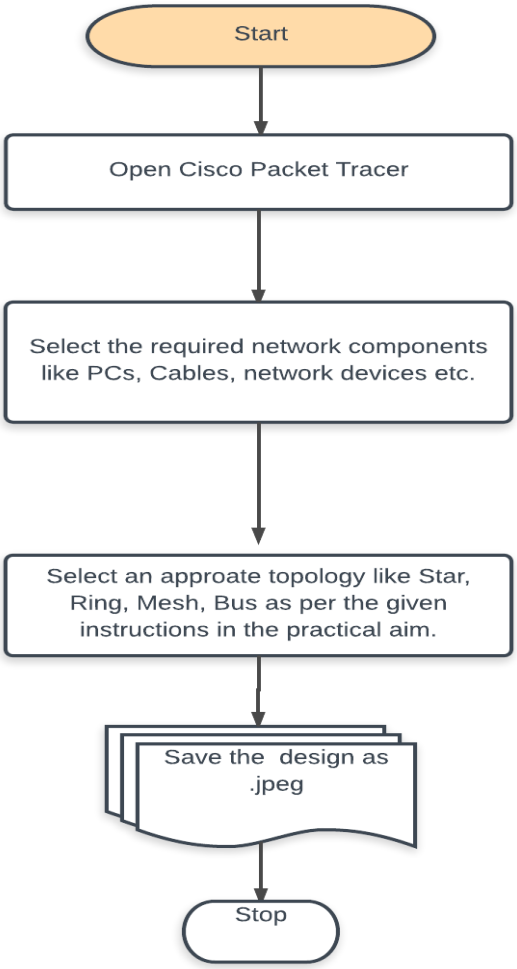
AIM: Internetworking with routers: Design a three or four simple networks (with 3 to 4 hosts) and connect via Router. Perform simulation and trace how routing is done in packet transmission.

1: Experiment on same subnet

2: Perform Experiment across the subnet and observe functioning of Router via selecting suitable pair of Source and destination.

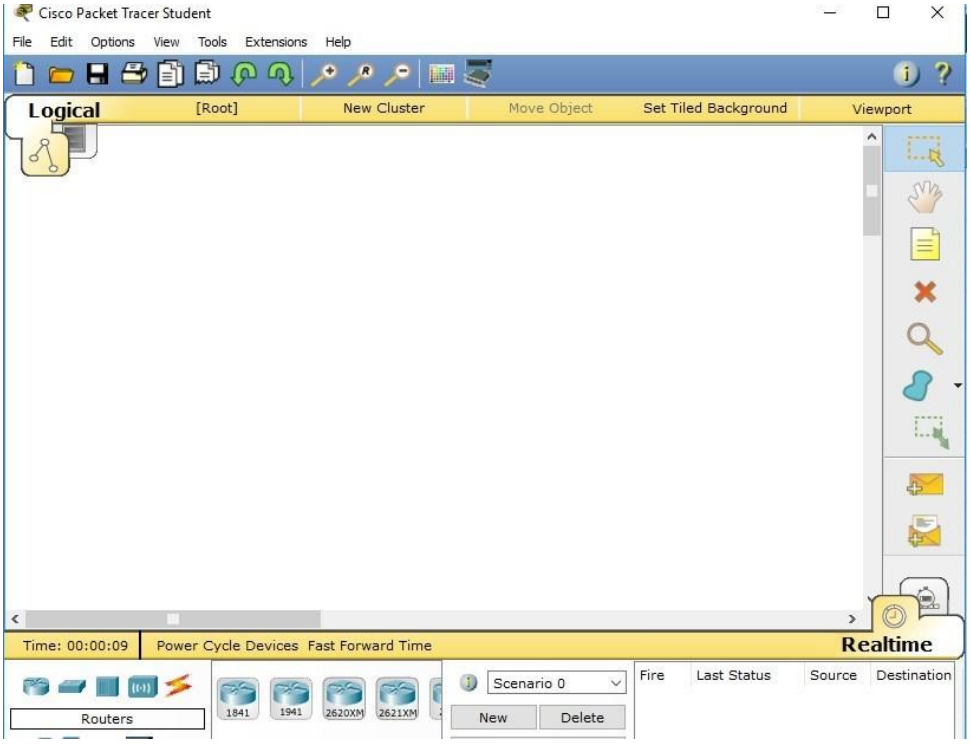
7.1 Purpose:
The main objective of the proposed experiment is to connect multiple networks using a router. It also gives an opportunity to students to learn router configuration in multiple network environments.

7.2 Logical Flow of Practical

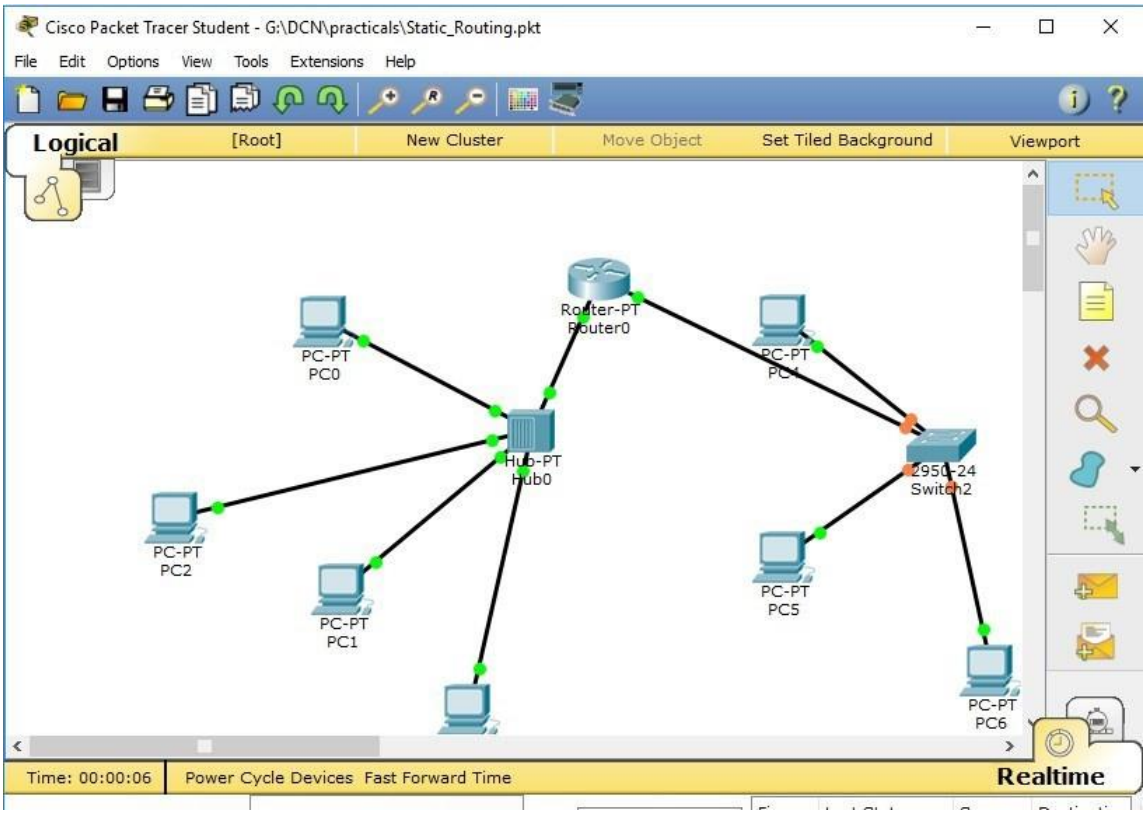


7.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



7.4 Expected Output:



7.5 Practice Excercises:

1. Design and connect multiple networks using a serial router connection.

2. Design and connect multiple networks using a parallel router connection

EXPERIMENT NO: 8

AIM: Implementation of SUBNETTING: Design multiple subnet with suitable number of hosts. Make a plan to assign static IP addressing across all subnet to explain implementation of SUBNETTING.

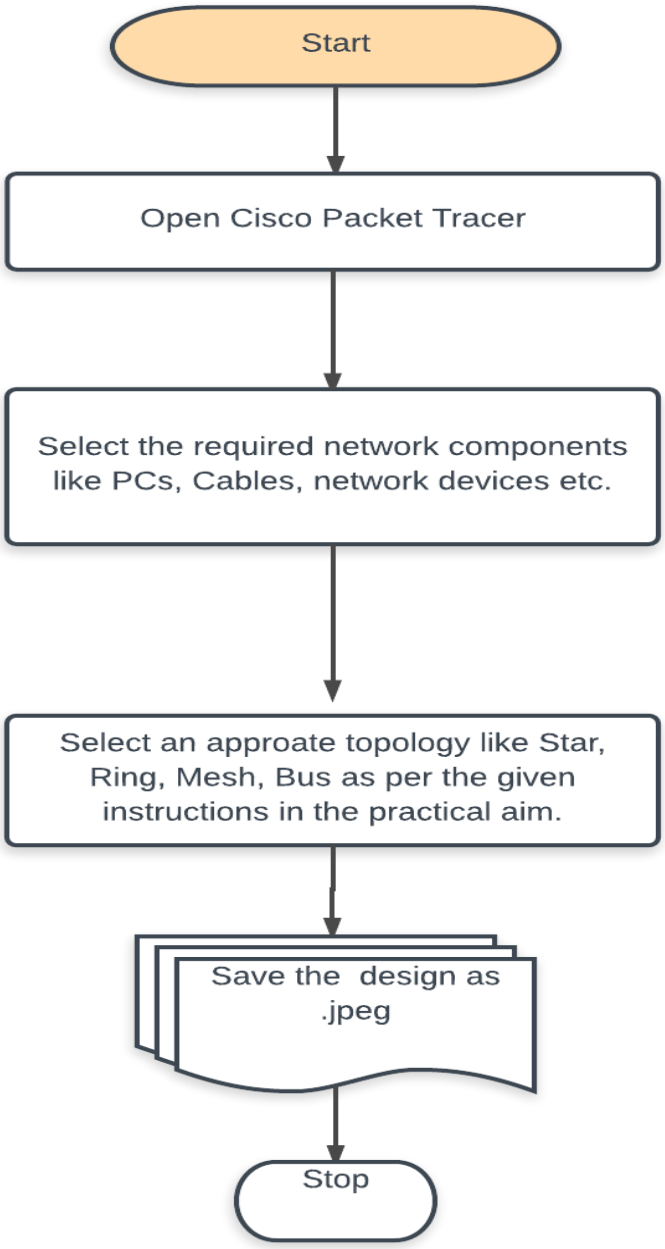
8.1 Purpose:
The main objective of the proposed experiment is to divide multiple networks using the concept of subnetting and super-netting.

8.2 Logical Flow of Practical

255.255.255.192 192 = 11000000

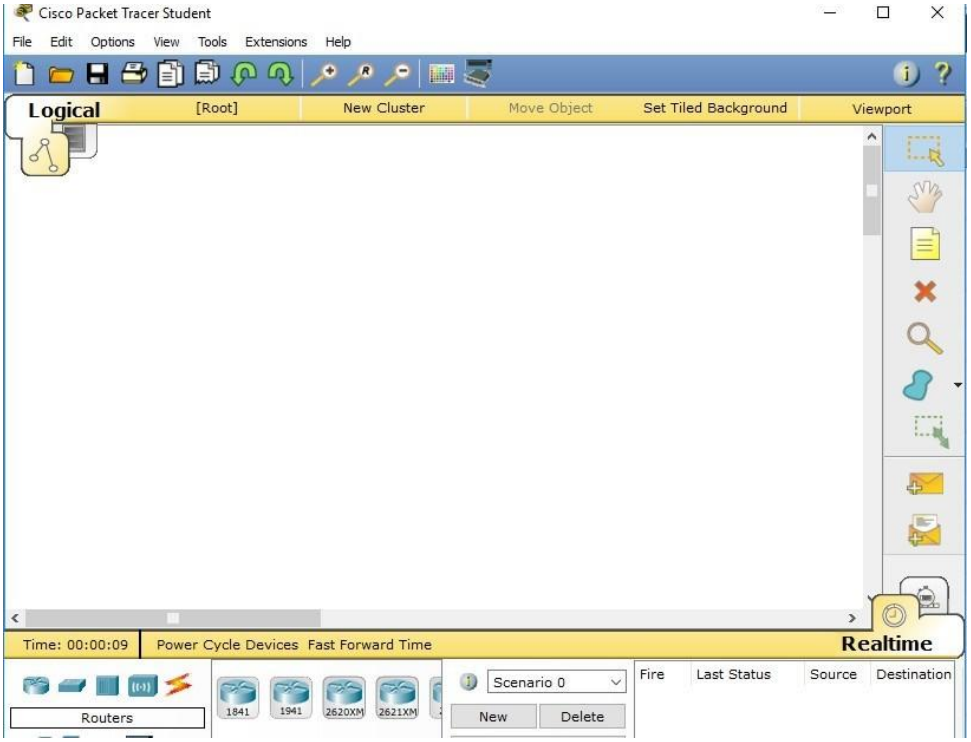
Subnets		Hosts					
1	1	0	0	0	0	0	0
128	64	32	16	8	4	2	1
2	1	32	16	8	4	2	1

subnets = 2 (4 - 2)
hosts = 62 (64 - 2)

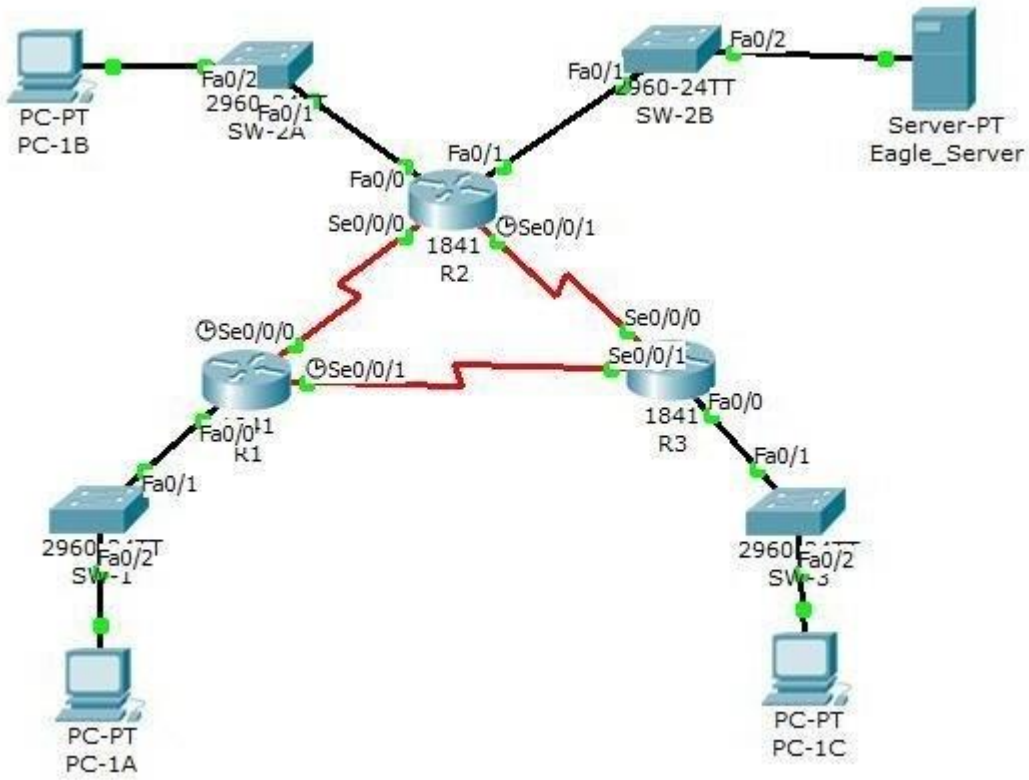


8.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



8.4 Expected Output:



8.5 Practice Excercises:

1. Design and connect multiple networks using a serial router connection.

2. Design and connect multiple networks using a parallel router connection

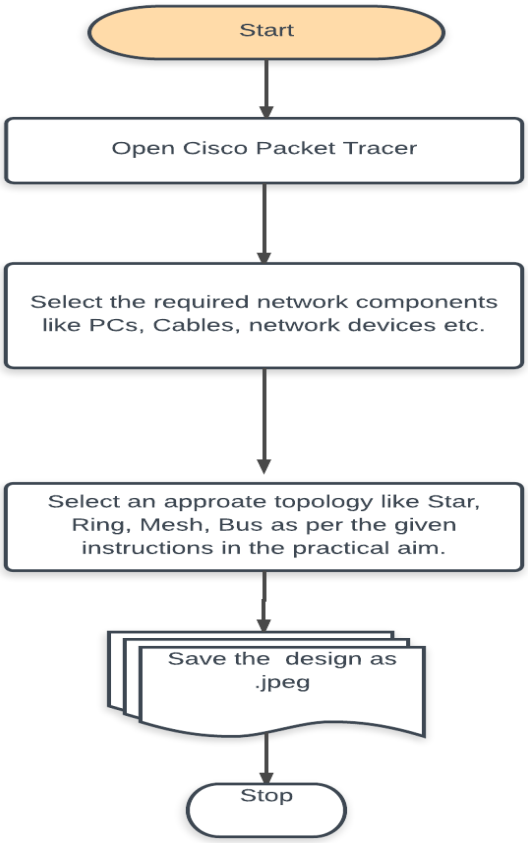
EXPERIMENT NO: 9

AIM: Routing at Network Layer: Simulate Static and Dynamic Routing Protocol Configuration using CISCO Packet Tracer.

9.1 Purpose:

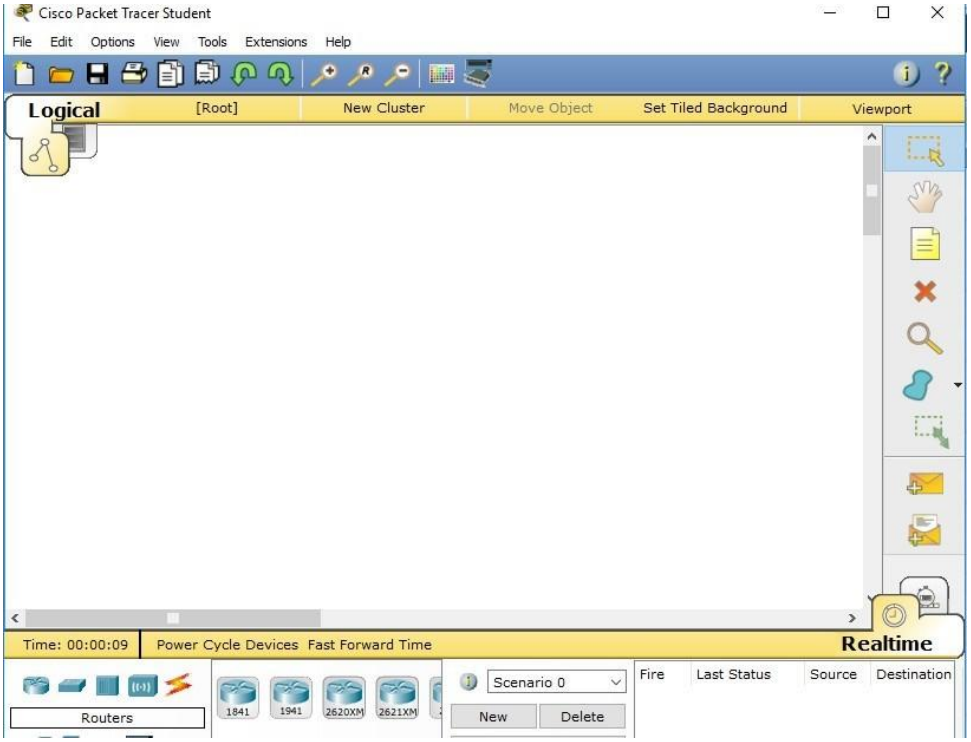
The main objective of the proposed experiment is to implement static and dynamic routing protocol using serial and parallel routing connections. It also gives an opportunity to students to learn serial and parallel router configuration in multiple network environments.

9.2 Logical Flow of Practical



9.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



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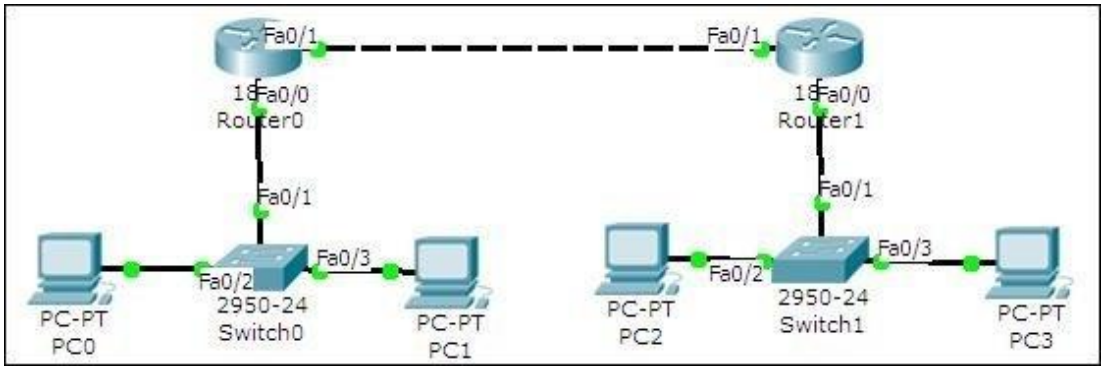
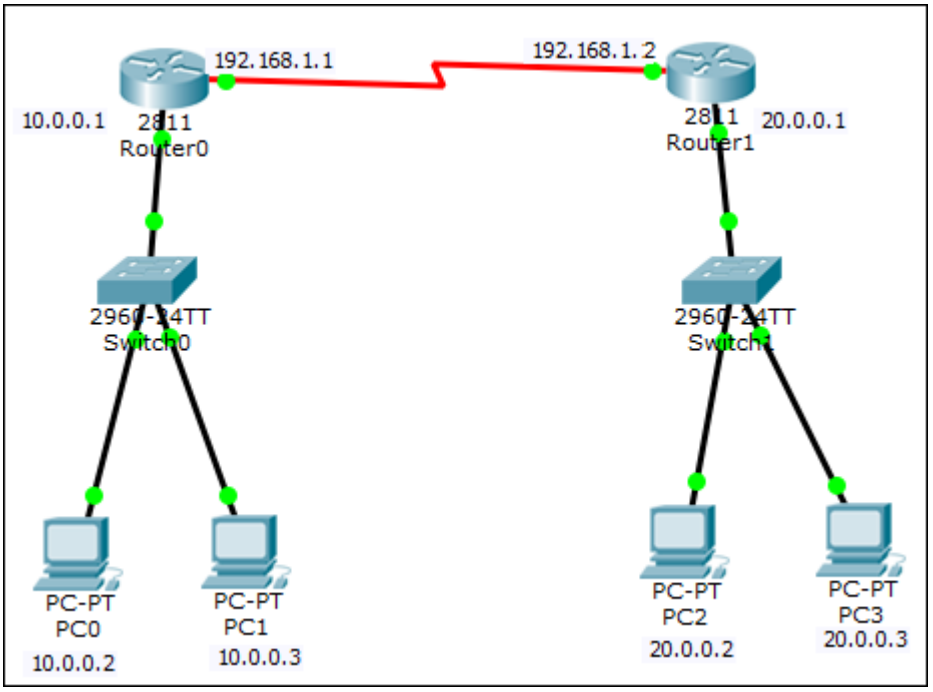




9.5 Practice Excercises:

1. Design an actual LAN network using 4 nodes. Design two subnets from the designed network using class A Ip address classes.

2. Design actual Wi-Fi network using 4 nodes. Design two subnets from the designed network using class A Ip address classes.

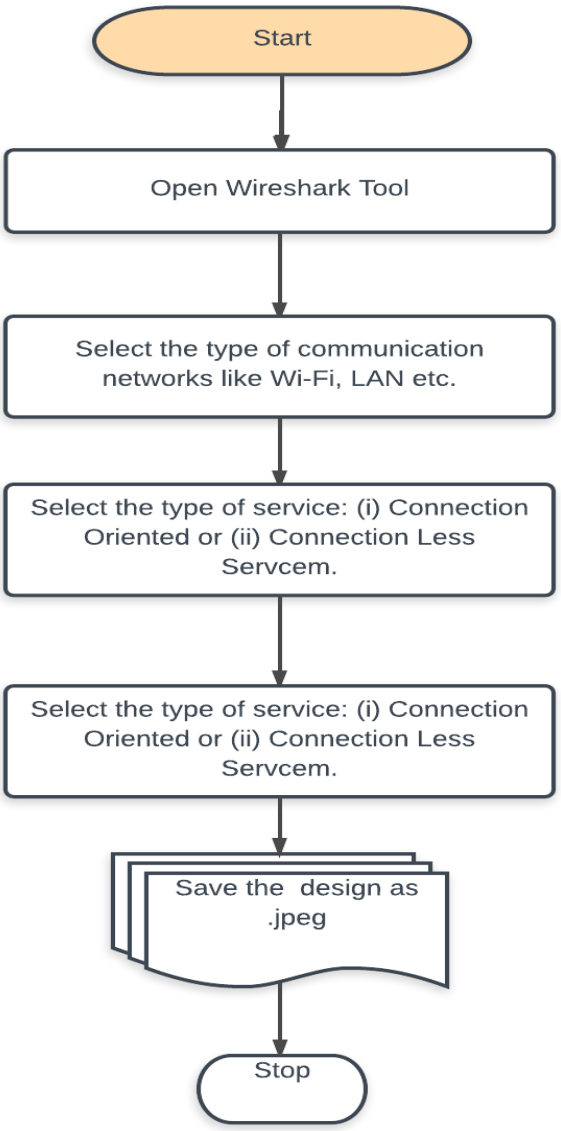
EXPERIMENT NO: 10

AIM: Experiment on Transport Layer: Implement echo client server using TCP/UDP sockets.

10.1 Purpose:

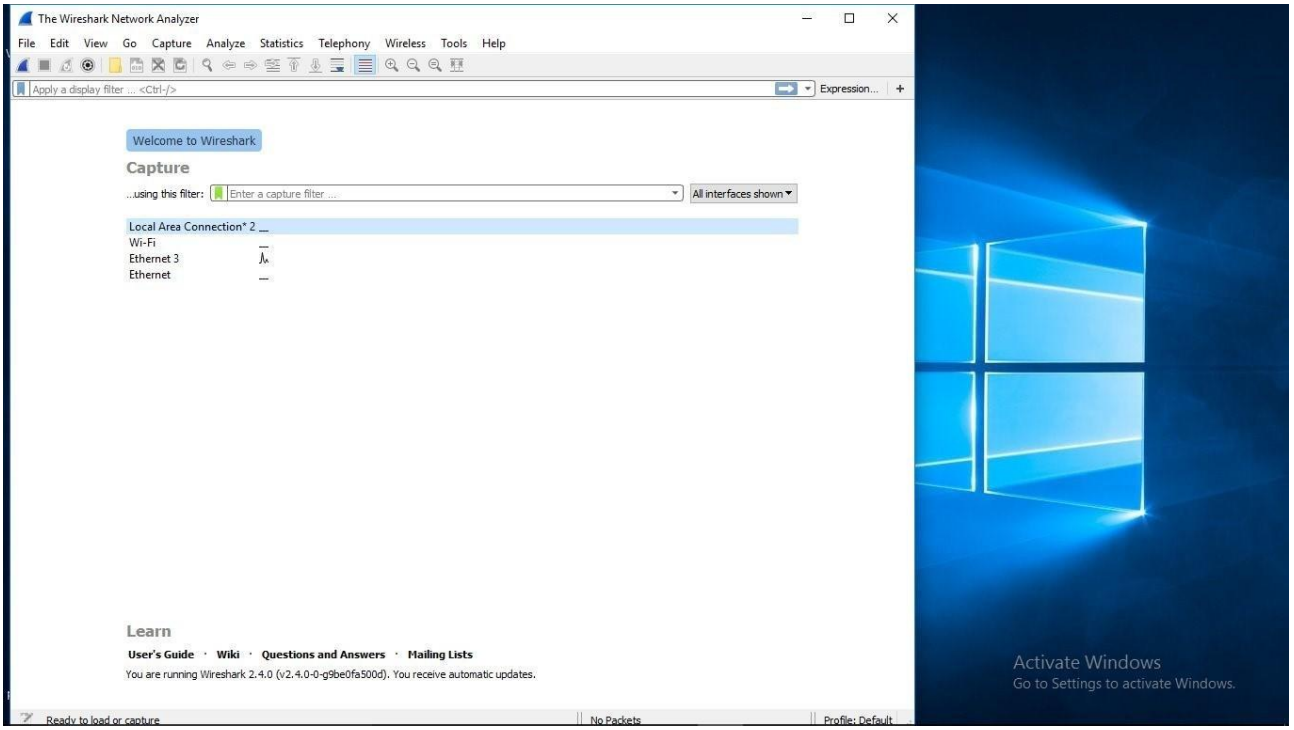
The main objective of the proposed experiment is to initiate communication between client and server nodes using echo client and echo server. Utilize both TCP and UDP sockets in the proposed experiment.

10.2 Logical Flow of Practical

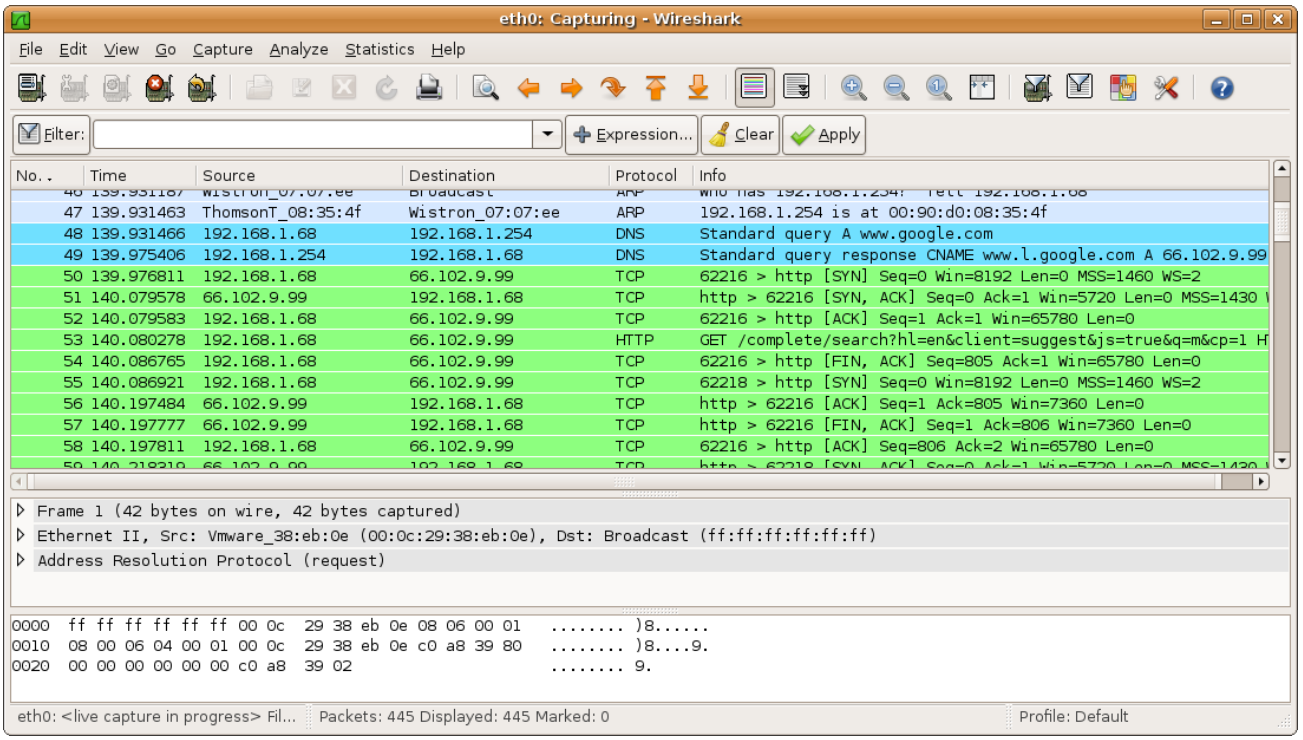


10.3 TOOLS/SOFTWARE:

Wireshark



10.4 Expected Output:



10.5 Practice Excercises:

1. **Test asynchronous and synchronous communication using wireshark in LAN environment.**
2. **Test asynchronous and synchronous communication using wireshark in Wi-Fi environment.**